ASSIGNMENT 3: GOOGLE CLOUD PLATFORM & BIG DATA - The Cloud Data Lake

Topics Covered: GCP Setup, Cloud Storage, IAM, Big Data Concepts, Data Lake Architecture

Duration: 10-12 hours | **Questions**: 30

Problem Statement

Design and implement a **Scalable Data Lake** on GCP for a financial services company. Handle structured, semi-structured, and unstructured data with proper governance.

Datasets

Public Financial & Economic Datasets

- > **Link 1** : https://www.kaggle.com/datasets/stefanoleone992/mutual-funds-and-etfs (Financial data)

- > **Link 2**: https://github.com/datasets/finance-vix (Market volatility data)

- > **Link 3**: https://www.sec.gov/dera/data/financial-statement-data-sets (SEC Financial Statements)

- **Use**: Upload these to GCS and build data lake architecture

GCP Free Tier: $300 credits for 90 days - enough for this assignment (no need for extra credits)

Assignment Questions (30 Total)

Part A: GCP Foundation (6 questions)

1. Create GCP project, enable billing, setup billing alerts (50%, 80%, 100% threshold)

2. Design IAM roles for: Data Engineers (read/write), Data Analysts (read-only), Admins (full access)

3. Create 3 Cloud Storage buckets with naming convention: `{project}-{env}-{layer}` (dev-bronze, dev-silver, dev-gold)

4. Implement bucket lifecycle policy: delete files in bronze after 90 days, move silver to Nearline after 30 days

5. Setup service account for programmatic access, generate JSON key, test with gsutil

6. Configure Cloud Logging and create alert for failed file uploads

 # Part B: Data Lake Architecture Design (8 questions)

7. Design 3-tier medallion architecture: Bronze (raw), Silver (cleaned), Gold (aggregated) - create folder structure

8. Explain when to use: Cloud Storage vs BigQuery vs Cloud SQL vs Firestore

9. Implement naming convention: `{source}_{entity}_{YYYYMMDD}_{version}.{ext}`

10. Design schema evolution strategy: how to handle new columns in incoming data?

11. Create data retention policy document: how long to keep data in each layer?

12. Design partitioning strategy: partition by date for time-series data

13. Compare file formats: CSV vs Parquet vs Avro - when to use what? Benchmark compression and query speed

14. Document data flow with diagram: Source → Bronze → Silver → Gold → Analytics

 # Part C: Big Data Concepts (8 questions)

15. Explain 5 Vs of Big Data with real examples from financial data (Volume, Velocity, Variety, Veracity, Value)

16. Compare: Data Lake vs Data Warehouse vs Data Lakehouse (create comparison table)

17. What is schema-on-read vs schema-on-write? Which fits data lake better? Why?

18. Explain Lambda Architecture: batch + speed + serving layers

19. Compare batch vs stream processing: when to use each for financial data?

20. Explain CAP theorem: Consistency, Availability, Partition tolerance - trade-offs in distributed systems

21. Design slowly changing dimension (SCD Type 2) strategy for customer data in data lake

22. What is Change Data Capture (CDC)? How to implement in data lake?

 # Part D: Hands-On Implementation (8 questions)

23. Upload 1GB+ financial dataset to GCS using gsutil with parallel uploads: `gsutil -m cp -r`

24. Create Cloud Function triggered on file upload to bronze layer - auto-move to silver after validation

25. Use `gcloud` CLI to create bucket, upload files, set permissions, list files

26. Create BigQuery external table pointing to Parquet files in GCS - query without loading

27. Setup Cloud Pub/Sub topic and subscription for event-driven data ingestion

28. Create Cloud Scheduler job to trigger daily data processing at 2 AM UTC

29. Implement data quality check: validate file schema before promoting bronze → silver

30. Build simple Looker Studio dashboard showing: files processed, data volume, success/failure rate

Deliverables

- Architecture diagram (Lucidchart/Draw.io) showing complete data flow
- GCP setup script (all gcloud commands in shell script)
- Data governance document (retention, access control, compliance)
- Cost estimation report (storage, compute, networking)
- 10-minute video walkthrough

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